Quantum-Centric Simulations of Extended Molecules: Combining QM/MM and DMET with Sample-Based Quantum Diagonalization

Gopal Ramesh Dahale[1]

[1] *Ecole Polytechnique Fédérale de Lausanne (EPFL), CH-1015 Lausanne, Switzerland*

Quantum embedding methods enable efficient molecular simulations by combining high-level quantum calculations on active regions with low-level treatments of the surrounding environment, leveraging both quantum and classical computing resources. Here, we present density matrix embedding theory (DMET) and quantum mechanics/molecular mechanics (QM/MM) simulations, both enhanced by sample-based quantum diagonalization (SQD). Using DMET-SQD with active space selection, we compute the ground-state energy of an H10 ring and relative energies of cyclohexane conformers (chair, half-chair, twist-boat, boat) achieving results consistent with classical methods. Additionally, QM/MM-SQD is applied to model the explicit solvation of glycine in water via electrostatic embedding. These DMET-SQD and QM/MM-SQD calculations demonstrate the potential of quantum-centric simulations for accurately treating large molecular systems with reduced computational cost.

I. INTRODUCTION

Simulating the electronic structure of large molecules, such as proteins, poses a major challenge for first-principles methods due to the high computational cost of beyond-mean-field approaches [1, 2]. For example, insulin simulations with correlation-consistent basis sets demand thousands of molecular orbitals, straining conventional methods [3–5]. Quantum embedding methods tackle this by partitioning large systems into smaller subsystems, applying high-level quantum calculations to active regions and lower-level methods to the environment [6–9]. These methods align well with quantum-centric supercomputing (QCSC), where quantum computers handle subsystem calculations. Classical high-performance computing manages environmental interactions [10]. The recently developed sample-based quantum diagonalization (SQD) method has enabled simulations of large active spaces, up to 77 qubits, for systems like metalloproteins and molecular dimers [11]. Here, we integrate SQD with density matrix embedding theory (DMET) [12] and quantum mechanics/molecular mechanics (QM/MM) [13] to simulate an H10 ring [14–17], cyclohexane conformers [18–21], and glycine solvation in water, using reduced qubit counts, showcasing efficient quantum-centric simulations of extended molecular systems. This work is inspired by [22] and follows a similar structure in presentation. We utilize open-source software Tangelo 0.4.3 [23] and extend it to incorporate SQD from Qiskit [24].

II. METHODS

Density Matrix Embedding Theory

(DMET) [25–29] is a wave-function-based quantum embedding method that enables high-level quantum calculations on molecular fragments while treating the environment with a mean-field approach. DMET partitions the D-dimensional Hilbert space of a quantum system into a small fragment (dimension $d_F \ll D$) and a larger environment (dimension $D_E = D - d_F$). The exact ground state $|\Psi\rangle$ is approximated via a Schmidt decomposition, reducing the problem to a $2d_F$-dimensional subsystem Hamiltonian $\hat{H}_{emb}$ acting on the fragment and its bath orbitals [25, 27]. In practice, DMET uses a mean-field Slater determinant for a system with N_{elec} electrons in N_{orb} spatial orbitals, dividing orbitals into fragment and environment sets. Diagonalizing the environment-environment block of the one-particle reduced density matrix yields bath orbitals, enabling the construction of $\hat{H}_{emb}$ as an active-space Hamiltonian [26, 27]:

$$\hat{H}_{emb} = \sum_{tu} \tilde{t}_{tu}\, \hat{E}^{t}_{u} + \sum_{tu,vw} \frac{(tu|vw)}{2}\, \hat{E}^{tv}_{uw}, \tag{1}$$

where tu span fragment and bath orbitals, and $\tilde{t}_{tu}$ incorporates environment contributions [27]. For multiple fragments, $\hat{H}_{emb}$ is solved for each, and a global chemical potential μ_{glob} is optimized in one-shot DMET to ensure the correct electron number [27].

Quantum Mechanics/Molecular Mechanics

Quantum Mechanics/Molecular Mechanics (QM/MM) [13] is a hybrid method that combines quantum mechanical (QM) calculations for a small, chemically active region with molecular mechanics (MM) forcefield-based modeling for the surrounding environment. This approach reduces computational cost while accurately capturing quantum effects, such as charge transfers and hydrogen bonds, critical for biochemical systems like proteins [30]. In QM/MM, the system is partitioned into a QM region, described by the Schrödinger equation, and an MM region, modeled classically, with interactions (e.g., electrostatic embedding) coupling the two.

Sample-Based Quantum Diagonalization

Sample-based quantum diagonalization (SQD) [11, 31, 32] is a quantum subspace method [33] that samples computational basis states $\chi = \{\mathbf{x}_1 \ldots \mathbf{x}_d\}$ from a quantum circuit $|\Phi_{\text{qc}}\rangle$ with probability $p(\mathbf{x}) = |\langle \mathbf{x}|\Phi_{\text{qc}}\rangle|^2$, solving the Schrödinger equation in the resulting subspace on a classical computer. Using the Jordan-Wigner mapping [34–36], these basis states represent Slater determinants, enabling efficient matrix element computation via Slater-Condon rules [11]. Here, we apply SQD to solve the DMET subsystem Hamiltonian (1) for an H10 ring and cyclohexane conformers, and to perform QM/MM simulations of glycine solvation in water, using a truncated local unitary cluster Jastrow (LUCJ) ansatz:

$$|\Phi_{\text{qc}}\rangle = e^{-\hat{K}_2} e^{\hat{K}_1} e^{i\hat{J}_1} e^{-\hat{K}_1} |\mathbf{x}_{\text{RHF}}\rangle, \tag{2}$$

where $\hat{K}_1$, $\hat{K}_2$ are one-body operators, $\hat{J}_1$ is a density-density operator, and $|\mathbf{x}_{\text{RHF}}\rangle$ is the restricted Hartree-Fock state, with parameters derived from classical CCSD [11]. On noisy quantum devices, the sampled distribution $\tilde{p}(\mathbf{x})$ may break particle-number and spin-z symmetries. We address this using the self-consistent configuration recovery (S-CORE) procedure [11], iteratively adjusting basis states based on the ground-state occupation number distribution $n_{p\sigma}$. For each iteration, K subspaces $S^{(b)}$ ($b = 1 \ldots K$) are constructed, and the Hamiltonian is projected as $\hat{H}_{S^{(b)}} = \hat{P}_{S^{(b)}} \hat{H} \hat{P}_{S^{(b)}}$, with $\hat{P}_{S^{(b)}} = \sum_{\mathbf{x} \in S^{(b)}} |\mathbf{x}\rangle\langle\mathbf{x}|$. The lowest energy across batches approximates the ground-state energy, updated iteratively via $n_{p\sigma}$ [11].

III. RESULTS

Results for N_2 with SQD

We applied sample-based quantum diagonalization (SQD) to compute the ground-state energy of N_2 using the cc-pVDZ basis with frozen core orbitals. Active space selection reduced the system from 26 orbitals and 10 electrons to 10 orbitals and 10 electrons, requiring 20 qubits. Using SQD with the LUCJ ansatz [37], we obtained an energy within 0.22mH Hartree of FCI, near chemical accuracy (1.6mH). Orbital occupancy analysis shows spin-up and spin-down electrons primarily occupy the first five orbitals. These results highlight SQD's accuracy for small molecules, supporting its use in DMET and QM/MM simulations.

DMET with SQD

Next, we applied DMET with SQD as well as VQE to butane and a ring of 10 hydrogen atoms (H10), using the minimal basis (minao) for butane and a second-quantized molecular framework for both systems. For butane, DMET with CCSD as the fragment solver, fragmenting the molecule into CH_3, CH_2, CH_2, CH_3 groups (4, 3, 3, 4 atoms), yields a correlation energy $E_{\text{corr}} = E_{\text{DMET}} - E_{\text{HF}} = 2.64 \times 10^{-1}$ Hartree, indicating significant electron correlation.

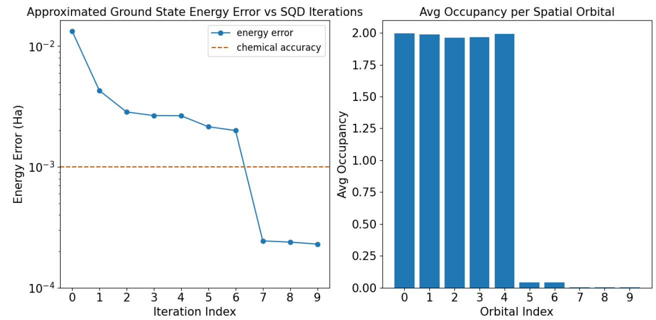

Figure 1. (left) Approximated Ground State Energy Error vs SQD Iterations and (right) Avg Occupancy per Spatial Orbital for N_2 with SQD

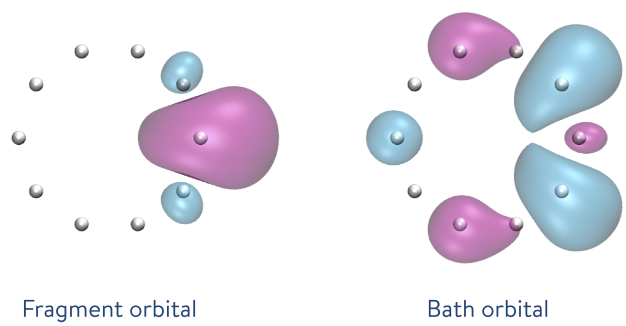

Figure 2. H10 Fragment and Bath. Courtesy Tangelo [23].

For the H10 ring, DMET decomposes the system into 10 single-hydrogen fragments, each with one fragment orbital and one bath orbital. Results are shown in Table I. These results demonstrate that DMET-SQD achieves near-FCI accuracy with reduced computational resources, enhancing its applicability to larger molecular systems within quantum-centric frameworks.

Table I. Relative energy differences for H10 using DMET with CCSD, VQE, and SQD solvers, reported in Hartree

Method	ΔE
$\Delta E_{\text{FCI-DMET-VQE}}$	1.34×10^{-2}
$\Delta E_{\text{FCI-DMET-SQD}}$	1.33×10^{-2}
$\Delta E_{\text{FCI-HF}}$	1.16×10^{-1}

DMET for Cyclohexane Conformations

We now use DMET with CCSD and SQD to compute the ground-state energies of cyclohexane conformations (chair, half-chair, twist-boat, boat) using the minimal basis (minao). DMET-CCSD fragments the molecule into chemically relevant groups, while DMET-SQD restricts the fragment active space (circuit width 4, depth 10) to reduce quantum resource demands. Table II reports relative energy differences with respect to the chair conformation for cyclohexane. DMET-SQD closely matches DMET-CCSD relative energies demonstrating near-CCSD accuracy with significantly fewer qubits, ideal for quantum-centric simulations.

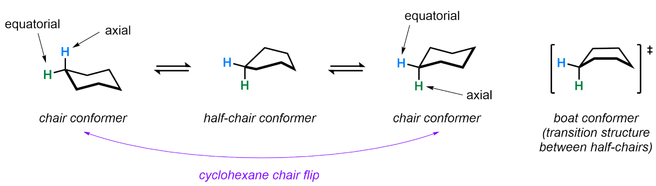

Figure 3. Cyclohexane Conformations [38]

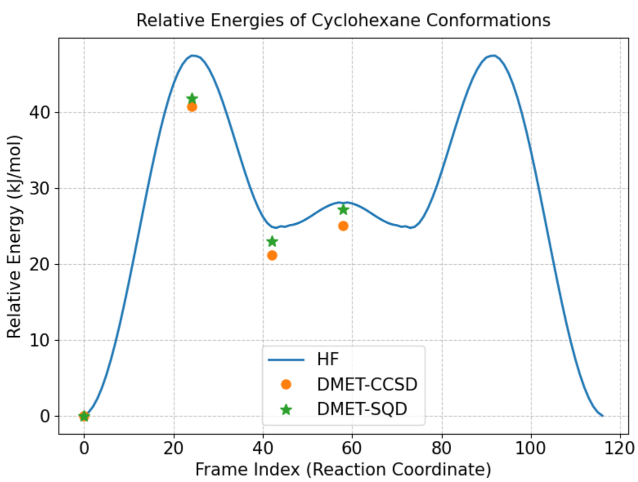

Figure 4. Relative Energies of Cyclohexane Conformations computed with Hartree-Fock, DMET-CCSD, and DMET-SQD.

Table II. Relative energy differences for cyclohexane conformations using DMET with CCSD and SQD solvers, in Hartree. Cyclohexane energies are relative to the chair conformation.

Conformation	$\Delta E_{\mathbf{SQD\text{-}CCSD}}$
Chair	4.53×10^{-1}
Half-chair	4.54×10^{-1}
Twist-boat	4.54×10^{-1}
Boat	4.54×10^{-1}

QM/MM with SQD

The QM/MM-SQD method was used to compute the hydration energies of neutral and zwitterionic glycine in a water cluster using electrostatic embedding. The results, summarized in Table III, show negative hydration energies, indicating stabilization of both tautomers in aqueous solution. The zwitterion form exhibits greater stabilization compared to the neutral form, attributable to enhanced electrostatic interactions with the polar water environment. These values align in order of magnitude with the experimental solvation enthalpy ($\Delta H_{\mathrm{solv}} \approx 122\,\mathrm{kJ/mol}$), despite the absence of entropic contributions in our calculations. The QM/MM-SQD approach effectively captures solvation effects, demonstrating its potential for efficient quantum-centric simulations of biochemical systems.

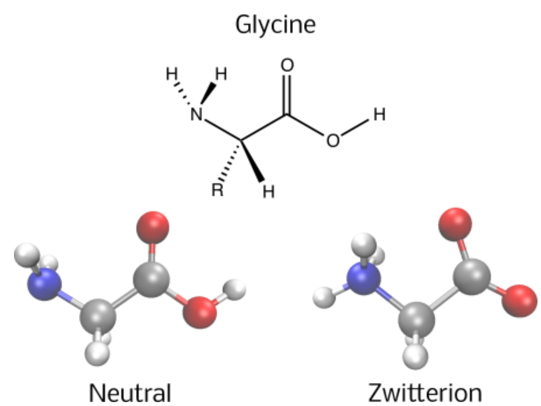

Figure 5. Tautomers of Glycine. Courtesy Tangelo [23].

Table III. Hydration energies (kJ/mol) of glycine tautomers computed using QM/MM-SQD.

Tautomer	**Hydration Energy**
Neutral	-136
Zwitterion	-291

IV. CONCLUSION

In this work, we present the integration of sample-based quantum diagonalization (SQD) with density matrix embedding theory (DMET) and quantum mechanics/molecular mechanics (QM/MM) to perform quantum-centric simulations of molecular systems. We applied DMET-SQD to compute the ground-state energy of an H10 ring and the relative energies of cyclohexane conformers, achieving near-CCSD accuracy with reduced qubit requirements. Additionally, QM/MM-SQD was employed to calculate the solvation energies of neutral and zwitterionic glycine, revealing enhanced stabilization of the zwitterion in aqueous environments, consistent with experimental trends. These calculations, leveraging SQD's noise-resilient properties [11], enable efficient treatment of larger subsystems than previously feasible on near-term quantum devices, supported by classical HPC resources for pre- and post-processing tasks. Our results underscore the potential of DMET-SQD and QM/MM-SQD within QCSC frameworks, paving the way for future studies targeting complex biochemical systems and non-minimal basis sets to enhance accuracy and applicability in molecular design.

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[1] D. A. Bardwell, C. S. Adjiman, Y. A. Arnautova, E. Bartashevich, S. X. Boerrigter, D. E. Braun, A. J. Cruz-Cabeza, G. M. Day, R. G. Della Valle, G. R. Desiraju, *et al.*, Towards crystal structure prediction of complex organic compounds–a report on the fifth blind test, Acta Crystallogr. B **67**, 535 (2011).

[2] J. Yang, W. Hu, D. Usvyat, D. Matthews, M. Schütz, and G. K.-L. Chan, Ab initio determination of the crystalline benzene lattice energy to sub-kilojoule/mole accuracy, Science **345**, 640 (2014).

[3] V. Timofeev, R. Chuprov-Netochin, V. Samigina, V. Bezuglov, K. Miroshnikov, and I. Kuranova, X-ray investigation of gene-engineered human insulin crystallized from a solution containing polysialic acid, Acta Crystallogr. F **66**, 259 (2010).

[4] Q. Sun, X. Zhang, S. Banerjee, P. Bao, M. Barbry, N. S. Blunt, N. A. Bogdanov, G. H. Booth, J. Chen, and Z.-H. Cui, Recent developments in the pyscf program package, J. Chem. Phys **153**, 024109 (2020).

[5] M. Valiev, E. J. Bylaska, N. Govind, K. Kowalski, T. P. Straatsma, H. J. J. Van Dam, D. Wang, J. Nieplocha, E. Aprà, T. L. Windus, *et al.*, Nwchem: A comprehensive and scalable open-source solution for large scale molecular simulations, Comp. Phys. Comm **181**, 1477 (2010).

[6] K. Kitaura, E. Ikeo, T. Asada, T. Nakano, and M. Uebayasi, Fragment molecular orbital method: an approximate computational method for large molecules, Chem. Phys. Lett **313**, 701 (1999).

[7] K. Raghavachari and A. Saha, Accurate composite and fragment-based quantum chemical models for large molecules, Chem. Rev **115**, 5643 (2015).

[8] D. A. Erlanson, B. J. Davis, and W. Jahnke, Fragment-based drug discovery: advancing fragments in the absence of crystal structures, Cell Chem. Bio **26**, 9 (2019).

[9] Q. Sun and G. K.-L. Chan, Quantum embedding theories, Acc. Chem. Res **49**, 2705 (2016).

[10] Y. Alexeev, M. Amsler, M. A. Barroca, S. Bassini, T. Battelle, D. Camps, D. Casanova, Y. J. Choi, F. T. Chong, C. Chung, *et al.*, Quantum-centric supercomputing for materials science: A perspective on challenges and future directions, Future Gener. Comp. Sys **160**, 666 (2024).

[11] J. Robledo-Moreno, M. Motta, H. Haas, A. Javadi-Abhari, P. Jurcevic, W. Kirby, S. Martiel, K. Sharma, S. Sharma, and T. Shirakawa, Chemistry beyond exact solutions on a quantum-centric supercomputer, arXiv:2405.05068 (2024), available at https://arxiv.org/abs/2405.05068.

[12] N. C. Rubin, A hybrid classical/quantum approach for large-scale studies of quantum systems with density matrix embedding theory, arXiv:1610.06910 (2016).

[13] A. Warshel and M. Levitt, Theoretical studies of enzymic reactions: Dielectric, electrostatic and steric stabilization of the carbonium ion in the reaction of lysozyme, Journal of Molecular Biology **103**, 227 (1976).

[14] J. Hachmann, W. Cardoen, and G. K. Chan, Multireference correlation in long molecules with the quadratic scaling density matrix renormalization group, J. Chem. Phys **125** (2006).

[15] L. Stella, C. Attaccalite, S. Sorella, and A. Rubio, Strong electronic correlation in the hydrogen chain: A variational monte carlo study, Phys. Rev. B **84**, 245117 (2011).

[16] M. Motta, D. M. Ceperley, G. K.-L. Chan, J. A. Gomez, E. Gull, S. Guo, C. A. Jiménez-Hoyos, T. N. Lan, J. Li, F. Ma, *et al.*, Towards the solution of the many-electron problem in real materials: Equation of state of the hydrogen chain with state-of-the-art many-body methods, Phys. Rev. X **7**, 031059 (2017).

[17] M. Motta, C. Genovese, F. Ma, Z.-H. Cui, R. Sawaya, G. K.-L. Chan, N. Chepiga, P. Helms, C. Jiménez-Hoyos, A. J. Millis, *et al.*, Ground-state properties of the hydrogen chain: dimerization, insulator-to-metal transition, and magnetic phases, Phys. Rev. X **10**, 031058 (2020).

[18] D. Barton and R. Cookson, The principles of conformational analysis, Chem. Soc. Rev **10**, 44 (1956).

[19] H. L. Strauss and H. M. Pickett, Conformational structure, energy, and inversion rates of cyclohexane and some related oxanes, J. Am. Chem. Soc **92**, 7281 (1970).

[20] D. A. Dixon and A. Komornicki, Ab initio conformational analysis of cyclohexane, J. Phys. Chem **94**, 5630 (1990).

[21] E. L. Eliel, Conformational analysis in heterocyclic systems: recent results and applications, Angew. Chem., Int. Ed. Engl **11**, 739 (1972).

[22] A. Shajan, D. Kaliakin, A. Mitra, J. Robledo Moreno, Z. Li, M. Motta, C. Johnson, A. A. Saki, S. Das, I. Sitdikov, *et al.*, Toward quantum-centric simulations of extended molecules: Sample-based quantum diagonalization enhanced with density matrix embedding theory, Journal of Chemical Theory and Computation **21**, 6801 (2025).

[23] V. Senicourt, J. Brown, A. Fleury, R. Day, E. Lloyd, M. P. Coons, K. Bieniasz, L. Huntington, A. J. Garza, S. Matsuura, R. Plesch, T. Yamazaki, and A. Zaribafiyan, Tangelo: An open-source python package for end-to-end chemistry workflows on quantum computers, arXiv:2206.12424 10.48550/arXiv.2206.12424 (2022), arXiv:2206.12424.

[24] A. A. Saki, S. Barison, B. Fuller, J. R. Garrison, J. R. Glick, C. Johnson, A. Mezzacapo, J. Robledo-Moreno, M. Rossmannek, P. Schweigert, I. Sitdikov, and K. J. Sung, Qiskit addon: sample-based quantum diagonalization (2024), accessed: 2024-11-14.

[25] G. Knizia and G. K.-L. Chan, Density matrix embedding: A simple alternative to dynamical mean-field theory, Phys. Rev. Lett **109**, 186404 (2012).

[26] G. Knizia and G. K.-L. Chan, Density matrix embedding: A strong-coupling quantum embedding theory, J. Chem. Theory Comput **9**, 1428 (2013).

[27] S. Wouters, C. A. Jiménez-Hoyos, Q. Sun, and G. K.-L. Chan, A practical guide to density matrix embedding theory in quantum chemistry, Journal of chemical theory and computation **12**, 2706 (2016).

[28] S. Wouters, C. A. Jiménez-Hoyos, and G. KL Chan, Five years of density matrix embedding theory, Fragmentation: toward accurate calculations on complex molecular systems , 227 (2017).

[29] H. Q. Pham, V. Bernales, and L. Gagliardi, Can density matrix embedding theory with the complete activate space self-consistent field solver describe single and double bond breaking in molecular systems?, J. Chem. Theory Comput **14**, 1960 (2018).

[30] M. J. Field, P. A. Bash, and M. Karplus, A combined quantum mechanical and molecular mechanical potential for molecular dynamics simulations, Journal of Computational Chemistry **11** (1990).

[31] K. Kanno, M. Kohda, R. Imai, S. Koh, K. Mitarai, W. Mizukami, and Y. O. Nakagawa, Quantum-selected configuration interaction: Classical diagonalization of hamiltonians in subspaces selected by quantum computers, arXiv:2302.11320 (2023), available at https://arxiv.org/abs/2302.11320.

[32] Y. O. Nakagawa, M. Kamoshita, W. Mizukami, S. Sudo, and Y.-y. Ohnishi, Adapt-qsci: Adaptive construction of input state for quantum-selected configuration interaction, arXiv:2311.01105 (2023), available at https://arxiv.org/abs/2311.01105.

[33] M. Motta, W. Kirby, I. Liepuoniute, K. J. Sung, J. Cohn, A. Mezzacapo, K. Klymko, N. Nguyen, N. Yoshioka, and J. E. Rice, Subspace methods for electronic structure simulations on quantum computers, Electron. Struct **6**, 013001 (2024).

[34] G. Ortiz, J. Gubernatis, E. Knill, and R. Laflamme, Simulating fermions on a quantum computer, Comp. Phys. Comm **146**, 302 (2002).

[35] R. Somma, G. Ortiz, J. E. Gubernatis, E. Knill, and R. Laflamme, Simulating physical phenomena by quantum networks, Phys. Rev. A **65**, 042323 (2002).

[36] R. D. Somma, *Quantum computation, complexity, and many-body physics*, Ph.D. thesis, Instituto Balseiro, S.C. de Bariloche, Argentina and Los Alamos National Labo- ratory, Los Alamos, USA (2005), arXiv:quant-ph/0512209.

[37] M. Motta, K. J. Sung, K. B. Whaley, M. Head-Gordon, and J. Shee, Bridging physical intuition and hardware efficiency for correlated electronic states: the local unitary cluster jastrow ansatz for electronic structure, Chem. Sci **14**, 11213 (2023).

[38] V. O. Chemistry, Cyclohexane conformers.